

Sequence representations

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can encode an ordered sequence with an explicit vocabulary.

I can use padding and a mask without inventing real events.

I can identify information lost by counts unknown symbols and truncation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Encode an ordered sequence with an explicit vocabulary

Use padding and a mask without inventing real events

Identify information lost by counts unknown symbols and truncation

Retrieve what you already know

Why can two different image regions have the same pooled maximum?

My starting explanation

This week's end goal

By the end, I can explain sequence representations using a traced example and a stated limitation.

2. Notice the evidence

Use fictional robot event logs whose symbols are GO, TURN and STOP. The task asks whether the final real event is STOP. This is a representation exercise, not a claim that an event log proves a robot is safe. GO then STOP and STOP then GO contain the same symbols but have different final events.

Log · Counts in GO TURN STOP order · Final event · Task answer

Log: GO STOP

Counts in GO TURN STOP order: [1, 0, 1]

Final event: STOP

Task answer: Yes

Log: STOP GO

Counts in GO TURN STOP order: [1, 0, 1]

Final event: GO

Task answer: No

Log: GO TURN STOP

Counts in GO TURN STOP order: [1, 1, 1]

Final event: STOP

Task answer: Yes

A count vector discards order, so no deterministic rule using only the first two count vectors can distinguish their answers. A sequence of IDs preserves order if we retain each position. Declare the vocabulary: PAD = 0, UNK = 1, GO = 2, TURN = 3 and STOP = 4. PAD is reserved for added storage positions; UNK represents a real but unrecognised symbol.

IDs are names, not measured amounts. STOP having ID 4 does not make it twice as much as GO with ID

2. Changing the vocabulary requires changing every consumer consistently. Do not interpret numerical distance between IDs as semantic similarity.

List the given facts and one fact that is not supplied.

3. See how it works

Store each log in four positions, padding on the right. A mask value 1 marks a real position and 0 marks padding. The encoder rejects empty logs, logs longer than four, and literal reserved markers. Other unrecognised symbols become UNK and still receive mask 1.

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")

    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")

    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)

    return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

The code assumes a list of symbol strings and uses exact spelling. It does not fit a vocabulary from data. If a future project fits vocabulary choices or frequency cutoffs, derive those choices from training data and apply the saved encoding consistently to validation and final cases.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: A missing sequence length causes every numeric0 to be treated as padding in the sensor example.

Prediction: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

5. Words that unlock the idea

Word · Meaning

Word: Sequence

Meaning: An ordered list of elements in which positions can matter

Word: Token ID

Meaning: A numerical identifier assigned to a symbol by a vocabulary

Word: Padding

Meaning: Placeholder positions added to make stored sequences the same length

Word: Mask

Meaning: A list marking which stored positions are real inputs and which are placeholders

Use two terms to explain a step in this week's example.

6. Follow a worked example

A padded log still has two real events

Stored position · Token ID · Mask

Stored position: 0

Token ID: 2 meaning GO

Mask: 1 real

Stored position: 1

Token ID: 4 meaning STOP

Mask: 1 real

Stored position: 2

Token ID: 0 meaning PAD

Mask: 0 placeholder

Stored position: 3

Token ID: 0 meaning PAD

Mask: 0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Encode TURN STOP, give its mask and real length, then identify the final real position and its

ID. Why would reading stored position 3 give the wrong event?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sequence representations

Distinguish padding positions from meaningful zero sensor readings.

The facts we will use

Toy tokens PAD0, UNK1, LEFT2, RIGHT3; stored length4. SequenceLEFT RIGHT encodes [2, 3, 0, 0], mask [1, 1, 0, 0].

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Sequence representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode real tokens

LEFT and RIGHT occupy positions 0 and 1.



2. Mark padding

Positions 2 and 3 are storage padding; mask marks them 0.



3. Compare sensor example

Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

CHANGE ONE THING

A missing sequence length causes every numeric 0 to be treated as padding in the sensor example.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Encode real tokens

What happens and why: LEFT and RIGHT occupy positions 0 and 1.

Step: 2. Mark padding

What happens and why: Positions 2 and 3 are storage padding; mask marks them 0.

Step: 3. Compare sensor example

What happens and why: Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Sequence values and sequence length are separate information. Padding fills unused storage positions so arrays can share a size; a mask marks which positions are real. A real sensor reading can be zero, just like a padding value. Keep the length or mask rather than inferring presence from the number alone. Token IDs are identifiers, not measured magnitudes.

What this example does not establish

Token padding rules need not apply to sensor values. Document sequence lengths, vocabulary and unknown-token handling.

Read the labelled picture: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: A missing sequence length causes every numeric0 to be treated as padding in the sensor example. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

A mask must affect a calculation to protect it from padding. Suppose real sensor values are [2, 4], stored as [2, 4, 0, 0] with mask [1, 1, 0, 0]. The masked mean is $(2 + 4) / 2 = 3$. Averaging all four slots gives 1.5. Unlike arbitrary token IDs, these sensor values are measurements for which a mean is meaningful.

Measured zero must remain real: values [0, 4, 0, 0] with mask [1, 1, 0, 0] have mean 2. Removing every zero would incorrectly give 4. The mask distinguishes a measured zero from a storage placeholder. An all-zero mask has no real values; reject it or follow an explicitly defined empty-input policy instead of dividing by zero.

Storing ordered IDs makes order available but does not guarantee a later model uses it. A computation that sums token counts or averages position-independent vectors can lose order again. Trace what the whole procedure reads, not just what the input format could preserve.

Locate the earliest unsupported step

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

State one repair and the checks you would rerun.

8. Practise together

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its mask, real length and final real event.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

Explain your prediction to a partner before comparing results.

Practise with less help: Expose an order collision

Use workbook W2 for the connected example “Sequence representations”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Sequence representations. Proposed explanation: “Larger token IDs mean more of a concept”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

```
VOCAB = {"GO": 2, "TURN": 3, "STOP": 4}

def encode(events):
    if not 1 <= len(events) <= 4:
        raise ValueError("Need one to four real events")
    if any(e in ("PAD", "UNK") for e in events):
        raise ValueError("Reserved markers are not events")
    ids = [VOCAB.get(e, 1) for e in events]
    n = len(ids)
```

```
return ids + [0] * (4 - n), [1] * n + [0] * (4 - n)

assert encode(["GO", "STOP"]) == ([2, 4, 0, 0], [1, 1, 0, 0])
assert encode(["WAIT", "STOP"]) == ([1, 4, 0, 0], [1, 1, 0, 0])
assert encode(["STOP", "GO"]) == ([4, 2, 0, 0], [1, 1, 0, 0])
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its mask, real length and

final real event.

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled sequence representations evidence sheet to Image-pattern Network Investigation. Show how you can encode an ordered sequence with an explicit vocabulary. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

WAIT STOP and PAUSE STOP both become [1, 4, 0, 0] with the same mask. UNK retains a real position but loses the identity of the unrecognised symbol. The final-event task can still answer Yes for both because the last real ID is STOP. A final real UNK would require an unknown result or another documented policy when the intended event cannot be determined.

A log GO TURN GO TURN STOP is cut to its first four events before encoding. What happens to the final-event answer? Explain why the lesson encoder rejects the original log instead of silently truncating it.

The full log ends in STOP, while the retained prefix ends in TURN. Truncation changes the evidence needed for this task. Save the original length and any truncation policy in a real experiment, and test the consequences. Next week introduces embeddings while preserving the distinction between identifiers and learned vectors.

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

A padded log still has two real events

Stored position · Token ID · Mask

Stored position: 0

Token ID: 2 meaning GO

Mask: 1 real

Stored position: 1

Token ID: 4 meaning STOP

Mask: 1 real

Stored position: 2

Token ID: 0 meaning PAD

Mask: 0 placeholder

Stored position: 3

Token ID: 0 meaning PAD

Mask: 0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a fresh schema independently

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its mask and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same mask. Calculate their masked mean and the incorrect mean over all five slots.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its mask and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same mask. Calculate their masked mean and the incorrect mean over all five slots.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to

demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Distinguish padding positions from meaningful zero sensor readings.

By the end, I can · How I will show it

By the end, I can: Encode an ordered sequence with an explicit vocabulary

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Use padding and a mask without inventing real events

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify information lost by counts unknown symbols and truncation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Encode an ordered sequence with an explicit vocabulary

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish padding positions from meaningful zero sensor readings.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

1. Encode real tokens
2. Compare sensor example
3. Mark padding

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare sensor example Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement. Sequence values and sequence length are separate information. Padding fills unused storage positions so arrays can share a size; a mask marks which positions are real. A real sensor reading can be zero, just like a padding value. Keep the length or mask rather than inferring presence from the number alone. Token IDs are identifiers, not measured magnitudes.

Practice 2: explain the deciding evidence

Changed-case practice: A missing sequence length causes every numeric0 to be treated as padding in the

sensor example. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The first real sensor reading is incorrectly discarded.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Repair a missing building block

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an input through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.